

The Stochastic Selector: Exact Rational Belief-State Safety Values under Partial Observation

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Abstract

The viable-selector framework reads every obstruction certificate as the emptiness of a response correspondence at an information state; its closing section previewed the stochastic counterpart, in which the information state is a distribution and viability is the value-one level of a belief-state safety recursion. This paper develops that theory. The belief-state safety value is the stochastic selector, and its finite-horizon values are piecewise linear with exactly representable witnesses: over rational input data, the backward recursion propagates rational alpha-vectors, so all values are exact rationals computable without floating point. At the deterministic limit the recursion degenerates to a survived-mass formula over blind action sequences, recovering the robust verdicts of the companion papers and implying their deficit bounds. Obstruction certificates are read systematically as deficit lower bounds: the chance-constrained common-action bound, the degenerate minimal-mass bound, and monotonicity. On the audited delayed-regime class the theory is carried exactly: the alpha-vector recursion reproduces the closed-form values on all forty-eight audited cells, the certificate partition coincides with the value boundaries, the value's kink locus is exactly the certificate boundary, the deficit is constant on the certified region, and the class declaration is quantified — the cells covered only by the timing certificate are precisely those where in-window adaptivity lifts the value to one. All computations run in exact rational arithmetic.

1. Introduction

The companion papers develop obstruction certificates for viability under incomplete observation — finite witnesses that no observation-based policy is safe — and unify them as emptiness of a response correspondence at an information state (Abaee, 2026a,b,c). The unifying paper closes with the stochastic counterpart, left there as an outlook: replace the set-valued information state by a distribution, the correspondence's nonemptiness by the value-one level of a safety recursion, and the certificate by a lower bound on the deficit from unity. This paper delivers that theory, seeded by the worked instance already carried in the calculus paper's probabilistic section and its coverage audit.

Four results. First, the *exact representation*: belief-state safety values are piecewise linear in the belief at every finite horizon, and over rational data the witnessing

alpha-vectors are rational and propagated exactly by the backward recursion (Section 3, Propositions 1 and 2) — no floating point enters the value theory, in keeping with the companions' verification discipline. Second, the *degeneration*: at the deterministic limit the recursion collapses to a survived-mass formula over blind action sequences, which recovers the companion papers' robust verdicts and proves their deficit bound (Proposition 2). Third, the *certificate reading*: obstruction certificates are deficit lower bounds, systematically (Proposition 3). Fourth, the *agreement theorem* on the audited classes: on the delayed hidden-regime grid the alpha-recursion, the closed form, the certificate partition, and the kink locus of the value coincide cell by cell, with a constant deficit on the certified region and a quantified class declaration (Section 5, Theorem 2). Everything is verified in exact rational arithmetic (Section).

2. The stochastic selector

Let X, A, Y be finite, with transition $T(x' | x, a)$ and observation likelihood $g(y | x, a, x')$. Augment X with an absorbing unsafe state $\perp$: for $x \in \mathcal{V}$, $T(\perp | x, a) = 1 - \sum_{x' \in \mathcal{V}} T(x' | x, a)$ and $T(\perp | \perp, a) = 1$. A belief is a distribution b over $X \cup \{\perp\}$, updated by Bayes' rule $\tau(b, a, y)$, with $\mathbb{P}(y | b, a) = \sum_{x, x'} b(x) T(x' | x, a) g(y | x, a, x')$. The *safety value* at horizon k is the maximal survival probability

$$V_k(b) = \max_{\pi} \mathbb{P}_{\pi}(x_t \in \mathcal{V} \text{ for } t = 0, \dots, k | b).$$

In the selector framework this is the distributional counterpart of the recursive correspondence: the information state is b , a response is a policy segment, and the *stochastic selector* at b ,

$$\Gamma_k^s(b) = \left\{ a \in A : \sum_y \mathbb{P}(y | b, a) V_{k-1}(\tau(b, a, y)) = V_k(b) \right\},$$

collects the actions attaining the value. Nonemptiness of the recursive selector corresponds to $V_k(b) = 1$; obstruction certificates correspond to deficits $1 - V_k(b) > 0$.

Theorem 1 (belief-state value recursion). $V_0(b) = b(\mathcal{V})$ (the horizon counts the current state), and for $k \geq 0$,

$$V_{k+1}(b) = \max_{a \in A} \sum_{y \in Y} \mathbb{P}(y | b, a) V_k(\tau(b, a, y)).$$

The recursion is exact on every finite horizon; it is the Smallwood–Sondik belief-state value iteration specialized to the absorbing safety reward (Smallwood and Sondik, 1973).

The proof is standard value iteration for the belief-state Markov decision process, with the absorbing state making V_k the k -step survival probability.

3. Exact piecewise-linear values

Proposition 1 (rational alpha-vectors). At every horizon k , V_k is piecewise linear and convex: $V_k(b) = \max_{\alpha \in \Gamma_k} \alpha^\top b$ for a finite witness set $\Gamma_k \subset \mathbb{Q}^{|X|+1}$, constructed by the backup

$$\alpha^{a, \gamma} (x) = \sum_{x'} T(x' | x, a) \sum_y g(y | x, a, x') \gamma_y(x'), \quad \gamma_y \in \Gamma_k,$$

with $\Gamma_0 = \{\mathbf{1}_Y\}$ and $\Gamma_{k+1} = \{\alpha^{a, \gamma} : a \in A, \gamma \text{ a selection}\}$, duplicates removed by exact equality. Over rational T , g , and priors, every $\alpha \in \Gamma_k$ is an exact rational vector, every $V_k(b)$ at a rational belief is an exact rational, and the attaining witness is part of the computation. The worst-case growth is $|\Gamma_{k+1}| \leq |A| |\Gamma_k|^{|Y|}$; deduplication and pruned selections are the practical controls, and the exponential law is the honest complexity statement for the general class.

Proof sketch. The PL structure is Smallwood–Sondik’s; the point here is the exact-arithmetic discipline: the backup maps rational vectors to rational vectors by finite sums and products, selections are finite, and equality tests for deduplication are exact on rationals. $\square$

Proposition 2 (deterministic-limit degeneration). Let the kernels be deterministic, $x^+ = F(x, a, d)$, with disturbance support $\{F(x, a, d) : d \in D\}$ read adversarially within its support, and let the window be blind with declared sequential class Π_B . Then the value degenerates to the *survived-mass formula*

$$V_k(b) = \max_{(u_0, \dots) \in \Pi_B} \sum_{x \in \text{supp } b} b(x) \mathbf{1}[x \text{ survives } (u_0, \dots)],$$

a maximum of rational linear functions of b (the surviving indicators are the degenerate alpha-vectors). Consequently: if the support B carries a jointly surviving blind sequence, $V_k(b) = 1$; and if every sequence loses some branch, then $V_k(b) \leq 1 - \min_{x \in B} b(x)$ — the robust common-action verdicts and the minimal-mass deficit bound of the companion papers, recovered as exact values.

Proof sketch. With deterministic kernels the posterior support follows the set-valued recursion, survival from a branch under a fixed sequence is a deterministic event, and prior mass sums over surviving branches; maximizing over sequences gives the formula, from which both consequences are immediate (a lost branch carries at least the minimal mass). $\square$

4. Certificates as deficit bounds

Write $\Delta_k(b) = 1 - V_k(b)$ for the deficit.

Proposition 3 (deficit lower bounds). Each companion certificate is a computable lower bound on the deficit: (i) *chance-constrained common action*: if the one-step violation probabilities satisfy $\sum_x b(x) p(x, a) \geq \delta$ for every $a \in A$, then $\Delta_k(b) \geq \delta$ for every $k \geq 1$; (ii) *degenerate minimal mass*: at the deterministic limit, if the support misses the k -step robust kernel, $\Delta_k(b) \geq \min_{x \in \text{supp } b} b(x)$; (iii) *monotonicity*: $V_{k+1}(b) \leq V_k(b)$, so the deficit is nondecreasing in the horizon. For finite horizons the exact deficit $1 - \max_{\alpha \in \Gamma_k} \alpha^\top b$ is computable, and the bounds are witnesses of it, not surrogates.

Proof sketch. (i) is the chance-constrained obstruction of the companion paper: every action loses probability at least δ at the first step and survival probabilities never increase. (ii) is Proposition 2’s second consequence. (iii) holds since $V_{k+1} = \max_a \mathbb{E}[V_k \circ \tau] \leq \max_a \mathbb{E}[V_{k-1} \circ \tau] = V_k$ by induction. $\square$

5. Agreement on the audited classes

The delayed hidden-regime instance of the calculus paper — regimes $\theta \in \{-1, +1\}$, blind window T_{obs} , declared hold class $\{u \equiv \pm 1\}$, prior $(\frac{1}{2}, \frac{1}{2})$ from z_0 — is the agreement test bed.

Theorem 2 (certificate–value agreement). On the audited grid ($z_0 \in \{1.0, \dots, 2.5\}$, $T_{\text{obs}} \in \{1, 2, 3\}$; 48 cells), over the declared hold class: (a) the alpha-vector recursion of Proposition 1 returns exactly the closed form

$$V_k(b_0) = \frac{1}{2} (\mathbf{1}[z_0 - k \geq 1] + \mathbf{1}[z_0 + k \geq 1]) \quad (k \leq T_{\text{obs}}),$$

$$V_k(b_0) = \frac{1}{2} (1 + \mathbf{1}[z_0 \geq 1 + T_{\text{obs}}]) \quad (k > T_{\text{obs}});$$

(b) the certificate partition coincides with the value boundaries: “viable” iff $V_k \equiv 1$; common-action-certified iff $V_1(b_0) < 1$; timing-certified iff $V_1(b_0) = 1$ but $V_{T_{\text{obs}}+1}(b_0) < 1$; (c) the attaining alpha-vector switches exactly at $z_0 = 2$: the value’s kink locus is the certificate boundary; (d) the deficit equals $\frac{1}{2}$ on every nonviable cell — the minimal-mass bound attained with the minimal prior; and (e) the declaration is quantified: over the unrestricted sequential blind class the value is 1 exactly on $\{z_0 \geq 2\}$ for $k \geq 2$ — the timing cells are precisely the cells where in-window adaptivity lifts the declared-class value.

Proof sketch. The two-regime blind POMDP has, at each window horizon, exactly two alpha-vectors $\alpha^u = (\mathbf{1}[z_0 - ut \geq 1 \ \forall t \leq \cdot], \mathbf{1}[z_0 + ut \geq 1 \ \forall t \leq \cdot])$, $u = \pm 1$; evaluating $\max_u \alpha^u b_0$ at the uniform prior gives (a), from which (b)–(d) read off against the companion paper’s certificate conditions, and (c) compares the attaining witness across the boundary. For (e), the deterministic degeneration (Proposition 2) over all alternating sequences keeps both branches within $z_0 \pm 1$. The complete computations are machine-verified in exact rational arithmetic. $\square$

6. Verified re-derivations

`paper2_stochastic_selector_v1_verify.py` (standard library; exact rational arithmetic; deterministic) re-derives every claim: *two-floor instance*: the backup produces exactly $\Gamma_1 = \{(1, 0), (0, 1)\}$, $V_k(b_0) = \frac{1}{2}$ for all probed horizons, the chance bound $\delta = \frac{1}{2}$ attained, and piecewise linearity confirmed at eleven simplex beliefs against the direct value $\max(b_1, b_2)$; *degeneration*: the exact survived-mass value equals brute-force evaluation at the probed beliefs, and at $(z_0, T) = (\frac{3}{2}, 2)$ every blind sequence loses a branch, so the deficit is at least the minimal mass; *agreement*: the recursion matches the closed form on all 48 cells and horizons $k \leq 4$; the partition coincidence holds cell by cell; the attaining witness switches at $z_0 = 2$ for every T_{obs} ; the deficit is exactly $\frac{1}{2}$ on all 42 nonviable cells; the unrestricted class attains value 1 exactly on $\{z_0 \geq 2\}$ for $k \geq 2$; *books*: 384 alpha-vectors generated across the campaign, all entries exact rationals (in this instance all in $\{0, 1\}$), and deficit monotonicity at every probed belief of both instances. Ten check families, all passing.

7. Delimitations and outlook

The semantics here is stochastic expectation: the value is a survival *probability*, and the robust quantifiers of the companion papers are recovered only at the deterministic degeneration (Proposition 2); the stochastic and robust readings of a genuinely stochastic plant differ, and this paper does not conflate them. The alpha-set growth is exponential in the worst case; the agreement theorem’s classes are two-witness instances precisely because the audited systems are. General information structures beyond history partitions (control-dependent observations, active sensing), continuous-time exit-time value functions, and the five-layer successor architecture are the next

items of the programme; the deficit-bound reading of Proposition 3 is the interface through which this theory attaches to the certificate families already established.

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Declarations

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Code availability. The verification script `paper2_stochastic_selector_v1_verify.py` (exact rational arithmetic; ten check families re-deriving Section) is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`).

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